

Pension Regime Choice under Informality:

Endogenous Sorting between PAYG and Funded Accounts

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July 2026

Abstract

This paper studies the joint choice of labor-market sector and pension regime in an economy with informality. I build a two-period overlapping-generations model in which heterogeneous workers choose among informal employment, formal employment with a funded account, and formal employment with a pay-as-you-go (PAYG) pension. Funded contributions are portable. PAYG eligibility increases with worker type, non-eligible contributors recover their notional balance, and the pension formula includes a floor and a ceiling. I discipline the Colombian calibration with legal parameters and long-run moments reported by the CEDE Pension Model. The total contribution rate is 16 percent; 11.5 percent is credited to a funded account and 13 percent to a PAYG notional account. The benchmark matches a 48.5 percent long-run formality anchor and an eligibility proxy derived from 25 percent contributory coverage. It produces 50.9 percent informality, 46.0 percent formal capitalization, and 3.1 percent formal PAYG. Workers below type 0.470 are informal, those between 0.470 and 0.771 choose capitalization, and higher types choose PAYG. PAYG accounts for 6.4 percent of formal workers, below the 10 percent enrollment share observed for young entrants. Easier PAYG eligibility raises participation but also raises the consumption tax and informality because the additional benefits are fiscally costly. The results establish a coherent mechanism for regime coexistence and a transparent calibration crosswalk; they are not an evaluation of the 2024 reform.

Keywords: pension choice; informality; funded accounts; pay-as-you-go; eligibility; overlapping generations. **JEL:** E21; H55; J26; J46.

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1 Introduction

Workers in economies with large informal sectors make two related decisions. They decide whether to participate in formal employment and, when the law permits it, which pension regime will receive their contributions. Treating pension assignment as exogenous suppresses the second margin. It can also conceal a corner solution: if the two formal regimes charge the same contribution rate and promise linear benefits, the regime with the higher implicit return attracts every formal worker.

This paper asks whether informality, funded accounts, and PAYG pensions can co-exist as endogenous choices. The model gives each worker three alternatives:

informal, formal–capitalization, formal–PAYG.

The pension regimes differ in three institutional dimensions. A funded account is portable and pays the accumulated market return. PAYG requires the worker to complete a contribution history, represented by an eligibility probability that increases with productivity. A worker who does not qualify recovers the notional balance without a real return. Conditional on qualifying, the PAYG pension follows a replacement formula with a floor and a ceiling.

These differences generate comparative advantage across worker types. Low types prefer informality because formal taxes and pension contributions exceed the value of formal-sector benefits. Middle types enter formal employment but choose capitalization because their PAYG eligibility remains low and their account is portable. High types choose PAYG because they are likely to qualify and receive its implicit subsidy. The model therefore produces two endogenous thresholds rather than imposing an assignment rule.

The benchmark equilibrium has the desired three-way sorting. Workers below type 0.470 are informal; workers between 0.470 and 0.771 choose capitalization; and workers above 0.771 choose PAYG. The corresponding population shares are 50.9, 46.0, and 3.1 percent. The consumption tax that balances the consolidated public budget is 19.6 percent.

The small PAYG share is economically informative. PAYG is selected only by high-productivity workers with high eligibility. Moreover, PAYG benefits exceed contemporaneous PAYG contributions in the calibration, so easier eligibility increases fiscal expenditure. When the eligibility intercept rises from -8.90 to -8.70 , the PAYG share

risks from 2.7 to 3.8 percent, but informality rises from 48.3 to 54.8 percent and the consumption tax from 17.1 to 23.7 percent. The general-equilibrium fiscal response reverses the partial equilibrium presumption that easier eligibility must increase formality.

The paper makes two contributions. First, it replaces exogenous pension assignment with a joint discrete choice that remains deterministic and interpretable. Second, it shows that eligibility and portability can produce regime coexistence without preference shocks. The calibration adds a third contribution: it distinguishes parameters that can be transferred from a Colombian microsimulation from those that cannot. Statutory contribution, account, and refund rates enter directly; long-run formality and contributory coverage discipline two parameters; the observed enrollment of young workers and the assumed real return are retained as external checks. The long-period conversion of PAYG benefits remains unidentified, so the regime shares should not be read as empirical predictions.

The rest of the paper reviews the related literature, presents the model and calibration, reports the equilibrium and sensitivity results, and describes the algorithm.

2 Related literature

The model follows the overlapping-generations approach to social security in [Diamond \(1977\)](#). Mandatory retirement programs can address undersaving ([Feldstein, 1985](#)), while present-biased preferences provide a behavioral rationale for mandatory contributions ([Laibson, 1998](#); [İmrohoroglu et al., 2003](#)). [Pardo \(2023\)](#) introduces mandatory retirement saving into an economy with endogenous informality. The present paper adds a second choice margin: formal workers select the institution that receives their contributions.

Pension design cannot be separated from labor informality. Payroll-financed programs alter the value of formal employment ([Levy, 2008](#)), and informality reflects choices by workers and firms rather than a residual labor-market state ([Ulyssea, 2018](#)). Search frictions and human-capital accumulation create heterogeneous formal careers ([Bobbà et al., 2021](#)). The shadow economy also changes the information and tax bases available for redistribution ([Doligalski and Rojas, 2023](#)).

Quantitative pension models with informality include [Joubert \(2015\)](#) and [McKiernan \(2020\)](#). Related work studies retirement and occupational behavior when formal contributions can be avoided ([Ferreira and Parente, 2020](#); [Moreno, 2022](#)). This paper focuses on a narrower mechanism: the interaction between pension eligibility, contribution portability, and the formal–informal margin. The objective is not to impose

observed regime shares, but to identify conditions under which the three alternatives are simultaneously optimal for positive masses of workers.

For Colombia, [Becerra \(2026\)](#) develops the CEDE Pension Model, a dynamic microsimulation that combines household surveys, administrative records, legal rules, and population projections. That model supplies the institutional rates and empirical anchors used below. Its annual partial-equilibrium accounting and the present model’s 40-year general-equilibrium period are not interchangeable. The calibration therefore uses a documented crosswalk rather than copying the entire parameter vector.

3 Model

3.1 Demography, heterogeneity, and preferences

One model period represents approximately 40 years. Population grows at rate n , labor-augmenting productivity at rate g , and a worker survives to old age with probability $1 - m$. Type $i \in [0, 1]$ has density

$$f(i) = \frac{2.4450}{1.006341403957648} \exp \left[- \left(\frac{i - 0.4655}{0.2329} \right)^2 \right], \quad \int_0^1 f(i) di = 1. \quad (1)$$

Sector-specific efficiency is

$$z_f(i) = \text{base}_f e^{\rho_f i}, \quad z_s(i) = \text{base}_s e^{\rho_s i}. \quad (2)$$

Instantaneous utility is CRRA, $u(c) = c^{1-\sigma}/(1-\sigma)$. Decision utility and experienced utility are

$$V_j(i) = u(c_{1j}(i)) + \beta(1 - m)\delta u(\tilde{c}_{2j}(i)), \quad (3)$$

$$W_j(i) = u(c_{1j}(i)) + (1 - m)\delta u(\tilde{c}_{2j}(i)), \quad (4)$$

where $j \in \{I, C, P\}$ denotes informal employment, capitalization, and PAYG. The parameter $\beta < 1$ affects saving and regime choice but is not used as a normative welfare weight.

3.2 Technology and household budgets

Formal and informal output are

$$Y_f = K^\alpha N_f^{1-\alpha}, \quad Y_s = A_s N_s. \quad (5)$$

Competitive pre-tax prices satisfy

$$\tilde{r} + \delta_k = \alpha(N_f/K)^{1-\alpha}, \quad \tilde{w}_f = (1-\alpha)(K/N_f)^\alpha, \quad \tilde{w}_s = A_s. \quad (6)$$

Taxes map these into the net return r and the effective wages w_f and w_s .

For choice j , voluntary saving $a_j(i) \geq 0$ satisfies

$$(1 + \tau_c)c_{1j}(i) + (1 + g)a_j(i) = y_{1j}(i), \quad (7)$$

$$(1 - m)(1 + \tau_c)c_{2j}(i) = (1 + r)a_j(i) + p_j(i). \quad (8)$$

Informal income is $y_{1I}(i) = w_s z_s(i)$. Both formal alternatives have

$$y_{1C}(i) = y_{1P}(i) = (1 - \tau_\ell - \tau_p) w_f z_f(i). \quad (9)$$

Both formal alternatives pay the statutory rate τ_p . The funded account credits $\tau_C \leq \tau_p$, while the PAYG notional balance credits $\tau_N \leq \tau_p$; uncredited amounts enter consolidated public revenue. The regimes therefore share the current wedge but differ in credited balances and retirement benefits. Baseline transfers are zero.

3.3 Capitalization

A funded contribution is fully vested and portable:

$$A_C(i) = \tau_C w_f z_f(i), \quad (10)$$

$$p_C(i) = \frac{(1 + r)A_C(i)}{1 + g}. \quad (11)$$

The contribution adds to the capital stock. The worker receives the accumulated value regardless of whether the subsequent formal career would have satisfied PAYG vesting requirements.

3.4 PAYG eligibility, refund, and pension formula

The probability that type i completes the PAYG eligibility requirement is

$$q(i) = \frac{\exp(\gamma_0 + \gamma_1 i)}{1 + \exp(\gamma_0 + \gamma_1 i)}, \quad \gamma_1 > 0. \quad (12)$$

The pension conditional on eligibility is

$$\mathcal{B}(y_i) = \min \{B_{\max}, \chi \max [B_{\min}, by_i]\}, \quad y_i = w_f z_f(i), \quad (13)$$

where b is the replacement parameter and χ converts the replacement flow into units of the long model period. It also captures the implicit subsidy embedded in the PAYG formula.

If the worker fails to qualify, a fraction φ of the notional balance is returned. Let r_N denote the annual real return credited to that balance. Expected PAYG retirement income is therefore

$$p_P(i) = q(i) \frac{(1 - m)\mathcal{B}(y_i)}{1 + g} + [1 - q(i)]\varphi \frac{(1 + r_N)^{40} \tau_N y_i}{1 + g}. \quad (14)$$

In the Colombian calibration, $\varphi = 1$ and $r_N = 0$: a non-eligible worker recovers the notional contribution adjusted for inflation but receives no real return. Expected PAYG benefits, including refunds, are public expenditure.

3.5 Joint choice and endogenous thresholds

Every worker solves the saving problem under all three alternatives and chooses

$$j^*(i) = \arg \max_{j \in \{I, C, P\}} V_j(i). \quad (15)$$

An interior three-regime equilibrium has thresholds $0 < i_1 < i_2 < 1$ satisfying

$$V_I(i_1) = V_C(i_1) > V_P(i_1), \quad (16)$$

$$V_C(i_2) = V_P(i_2) > V_I(i_2). \quad (17)$$

The resulting allocation is

$$j^*(i) = \begin{cases} I, & i < i_1, \\ C, & i_1 \leq i < i_2, \\ P, & i \geq i_2. \end{cases} \quad (18)$$

Equation (18) is an equilibrium outcome. The algorithm evaluates all three utilities point by point and does not impose either threshold.

3.6 Aggregation and fiscal closure

Let $d_j(i) = \mathbf{1}\{j^*(i) = j\}$. Regime shares and effective labor are

$$\pi_j = \int_0^1 d_j(i) f(i) \, di, \quad (19)$$

$$N_f = \int_0^1 [d_C(i) + d_P(i)] z_f(i) f(i) \, di, \quad (20)$$

$$N_s = \int_0^1 d_I(i) z_s(i) f(i) \, di. \quad (21)$$

Next period's capital is

$$K' = \frac{1}{1+n} \int_0^1 a_{j^*(i)}(i) f(i) \, di + \frac{1}{(1+n)(1+g)} \int_0^1 d_C(i) \tau_C w_f z_f(i) f(i) \, di. \quad (22)$$

Public pension revenue is

$$R^{pen} = \int_0^1 [d_P(i) \tau_p + d_C(i) (\tau_p - \tau_C)] w_f z_f(i) f(i) \, di. \quad (23)$$

With expected PAYG benefits B^{pg} , the consolidated public budget is

$$\tau_c \mathcal{C} + \tau_\ell w_f N_f + \tau_n w_f N_f + \tau_k \tilde{r} K + R^{pen} = G + \mathcal{T} + B^{pg}. \quad (24)$$

The consumption tax τ_c adjusts to satisfy (24). Consequently, a policy that raises expected PAYG benefits can affect all three choices through the fiscal response.

4 Calibration

4.1 Parameter crosswalk

The Colombian inputs come from the CEDE Pension Model documented by [Becerra \(2026\)](#). That source is an annual dynamic microsimulation; the present model is a stationary 40-year OLG equilibrium. I therefore classify each input as a direct legal parameter, a comparable calibration moment, an external validation, or a retained structural parameter. This prevents annual accounting assumptions from being reinterpreted as balanced-growth parameters.

The direct legal inputs are the 16 percent total contribution rate, the 11.5 percent credited to the individual retirement account, and the 13 percent credited to the PAYG notional account. A funded worker pays the full 16 percent: the 11.5 percent account balance becomes capital and the remaining 4.5 percent enters consolidated public revenue, which parsimoniously combines the minimum-pension fund, administration, and insurance components not separately modeled here. A PAYG worker also pays 16 percent, while a failed eligibility draw returns the 13 percent notional balance with no real interest. The statutory DB replacement formula is approximately 65 percent at one minimum wage, so I set $b = 0.65$.

Table 1 reports the complete benchmark. Annual population growth, productivity growth, and discounting remain compounded over 40 years: $n = 1.013^{40} - 1$, $g = 1.004^{40} - 1$, and $\beta = 0.97^{40}$. The source assumes 1.6 percent annual real wage growth, but that rate only indexes earnings and benefits in its microsimulation. Substituting it for balanced-growth productivity g changes every intergenerational scaling equation and does not yield a stable benchmark. Likewise, the source’s 4 percent real return is retained as an external check rather than imposed on the endogenous return of a closed economy.

4.2 Moments and external checks

The CEDE projection raises formal employment from 42 percent in 2020 to 48.5 percent in 2100. I use 48.5 percent as the stationary formality anchor. It also projects contributory coverage near 25 percent in 2080–2100. Because the OLG has no contribution-history state, I approximate conditional eligibility among formal workers by $0.25/0.485 = 0.51546$. This ratio is a calibration proxy, not an accounting identity: contributory coverage in the source includes DB, DC, and DB+DC pensions and is measured among

Table 1: Baseline parameters and provenance

Parameter	Value	Use	Interpretation
m	0.4000	retained	mortality before old age
n	0.6764	retained	population growth per model period
σ	2.5000	retained	CRRA curvature
β	0.2957	retained	behavioral discount factor
g	0.1731	retained	balanced-growth productivity
α	0.4000	retained	capital share
A_s	0.08215	calibrated	matches the long-run formality anchor
ρ_s, ρ_f	2.90, 4.64	retained	type gradients
τ_ℓ, τ_n	0.125, 0.125	retained	labor and payroll rates
τ_p	0.1600	direct	total pension contribution
τ_C	0.1150	direct	credited funded-account rate
τ_N	0.1300	direct	credited PAYG notional rate
r_N	0.0000	direct	real return on a PAYG refund
τ_k	0.3200	retained	capital-income tax
b	0.6500	direct	replacement rate near one minimum wage
G	0.3300	retained	government purchases
γ_0	-8.7900	calibrated	matches the eligibility proxy
γ_1	15.0000	retained	shape normalization for eligibility
φ	1.0000	direct	refund fraction if not eligible
χ	2.9000	selected	40-year benefit conversion; interior branch
$B_{\min}$	0.0500	retained	floor in internal OLG units
$B_{\max}$	100.0000	retained	nonbinding ceiling

older adults.¹

The two comparable moments discipline A_s and γ_0 . The 10 percent DB enrollment share among young entrants and the 4 percent real return in the source are not imposed. They are external checks because the former is a cohort-specific plan assignment and the latter is an actuarial assumption rather than a general-equilibrium price. Table 2 reports both the matched moments and these transparent discrepancies.

Table 2: Calibration moments and external validation

Use	Moment	Target	Model	Gap
Calibrated	Formal share	0.48500	0.49081	0.00581
Calibrated	Eligibility among formals	0.51546	0.51373	-0.00173
Validation	PAYG share among formals	0.10000	0.06353	-0.03647
Validation	Annual real return	0.04000	0.05741	0.01741

Notes: The eligibility target is the proxy 0.25/0.485. The PAYG validation uses the approximately 10 percent DB enrollment share among entrants aged 20–30 reported for 2017.

The source does not identify the logistic slope, the conversion from an annual pension

¹The coverage rows in Table 2 of [Becerra \(2026\)](#) are labeled in the reverse order. Section 4.1, Figure 4, and Table 1 consistently report contributory coverage of 23.7–23.8 percent on average and non-contributory coverage of 53.1–53.3 percent. The calibration uses the separately reported long-run contributory value of approximately 25 percent.

flow to a 40-year benefit, or the minimum wage in the model’s efficiency-wage units. Accordingly, γ_1 , χ , and $B_{\min}$ are not claimed as empirical estimates. The value $\chi = 2.9$ selects a locally stable interior branch. The benchmark is solved first on a 101-type grid and then refined on a 501-type grid. Its maximum fixed-point residual is 1.8×10^{-7} .

5 Results

5.1 Benchmark equilibrium

Table 3 reports the refined stationary equilibrium. All three alternatives have positive mass. Informality represents 50.9 percent of workers, capitalization 46.0 percent, and PAYG 3.1 percent. PAYG therefore accounts for 6.4 percent of formal workers. The formal-entry threshold is $i_1 = 0.4696$ and the pension-regime threshold is $i_2 = 0.7713$.

Table 3: Endogenous-choice steady state

Outcome	Value	Outcome	Value
Informal share	0.50919	Capital stock	0.02968
Capitalization share	0.45962	Formal labor	8.89024
PAYG share	0.03118	Informal labor	1.40926
First threshold i_1	0.46965	Consumption tax	0.19612
Second threshold i_2	0.77128	Annualized return	0.05741
Young consumption	0.39523	Formal wage	0.05450
Old-age consumption	0.27038	Voluntary capital	0.00620
Funded capital	0.02348	PAYG contributions	0.01328
Decision welfare	−5.43803	PAYG benefits	0.04661
Experienced welfare	−5.93741	Maximum residual	1.8×10^{-7}

The first threshold reflects the formal tax wedge. Types below 0.4696 obtain too little formal-sector income to compensate for labor and pension contributions. Above that threshold, formal productivity grows faster than informal productivity and capitalization becomes optimal. PAYG does not attract the middle group because its probability of completing the eligibility requirement is still too low.

The second threshold reflects the interaction between eligibility and the implicit PAYG subsidy. At $i = 0.7713$, the expected PAYG benefit overtakes the portable funded balance. Higher types remain in PAYG because eligibility continues to increase and the benefit ceiling does not bind. The two thresholds therefore arise from different comparisons: $V_C - V_I$ at the first threshold and $V_P - V_C$ at the second.

PAYG benefits are 3.51 times contemporaneous PAYG contributions. Part of this difference reflects demographic scaling and part reflects χ , which is required to compare an annual replacement flow with a 40-year funded balance. The consolidated budget finances the gap. This is why regime choice cannot be analyzed independently of fiscal closure. The model’s 6.4 percent PAYG share among formals is below the 10 percent enrollment check from young entrants. Closing that gap mechanically by raising χ moves the equilibrium toward a high-tax PAYG branch, rather than providing an innocuous fit.

5.2 Eligibility sensitivity

Table 4 changes only the eligibility intercept. Increasing γ_0 makes every type more likely to qualify while preserving the slope of the profile. Positive masses remain in all three alternatives throughout the reported range.

Table 4: Sensitivity to PAYG eligibility

	$\gamma_0 = -8.90$	Benchmark	$\gamma_0 = -8.70$
Informal share	0.48269	0.50929	0.54806
Capitalization share	0.49050	0.45951	0.41403
PAYG share	0.02681	0.03120	0.03792
Consumption tax	0.17123	0.19622	0.23667
First threshold i_1	0.45874	0.46969	0.48570
Second threshold i_2	0.78201	0.77125	0.75696

Notes: Sensitivity equilibria use 201 types and the refined benchmark as their initial state. All three converge and have maximum fixed-point residuals below 4.1×10^{-6} .

Easier eligibility lowers the second threshold and raises PAYG participation, as expected. It also increases expected benefits and the consumption tax. The higher tax raises the first threshold, so informality increases by 6.5 percentage points between the two endpoints. The capitalization share falls by 7.6 points: some workers enter PAYG while others leave formality. This is a general-equilibrium implication of the calibrated fiscal rule, not a claim that easier eligibility must raise informality in every pension system.

6 Conclusion

Informality, capitalization, and PAYG can coexist without exogenous pension assignment. The model generates two endogenous thresholds because funded balances are portable while PAYG benefits depend on eligibility and include an implicit subsidy. Low types choose informality, middle types choose formal capitalization, and high types choose formal PAYG.

The Colombian crosswalk sharpens the quantitative exercise. Statutory contribution, account-crediting, refund, and replacement rates now come from an explicit institutional source. The benchmark approximates long-run formality and the contributory-eligibility proxy while preserving an interior allocation: 50.9 percent informal, 46.0 percent capitalization, and 3.1 percent PAYG. Local changes in eligibility preserve all three alternatives. They also reveal the fiscal tension: easier PAYG access increases expected benefits, the consumption tax, and informality.

The audit also identifies what the calibration cannot support. The source's annual wage growth is not the model's balanced-growth productivity rate, and its assumed 4 percent actuarial return is not an endogenous closed-economy price. The 10 percent DB enrollment of young entrants is a useful external check, but the model produces 6.4 percent PAYG among formals. Forcing that fit through the benefit multiplier pushes the economy toward a high-tax branch. Treating non-comparable quantities as direct parameters would therefore create spurious precision.

The remaining magnitudes are provisional. Eligibility still depends only on productivity; the model has no separate contribution-history state, sex, or education. The next empirical step is to import the contribution densities behind the CEDE transitions, map the minimum wage into efficiency-wage units, and value statutory DB and DC benefits with the same annuity factor. Those additions would permit a genuine reform experiment. The present stationary competition between PAYG and capitalization is not a simulation of Colombia's 2024 complementarity reform. Random-utility shocks are not required for the interior equilibrium reported here.

A Numerical algorithm

A.1 Household alternatives

For each candidate aggregate state (K, N_f, τ_c) , the algorithm computes prices and solves three household problems at every diagnostic type:

1. informal employment with no pension contribution;
2. formal employment with a fully funded contribution;
3. formal employment with PAYG eligibility, pension, and refund given by (12)–(14).

The saving solution imposes nonnegative voluntary saving and positive consumption. The algorithm records all three decision utilities and applies (15).

A.2 Locating endogenous thresholds

The diagnostic grid identifies every interval in which the selected alternative changes. Within each interval, a scalar root finder solves the relevant pairwise utility equation. The roots are added to the quadrature nodes. Choice is then evaluated at each interval midpoint and held constant only within that root-bounded interval. This procedure neither assumes the number of crossings nor imposes their order.

A.3 General-equilibrium fixed point

Given the exact choice intervals, trapezoidal quadrature aggregates labor, consumption, voluntary saving, funded contributions, PAYG flows, and welfare. The update map is

$$(K, N_f, N_s, \tau_c) \mapsto (\widehat{K}, \widehat{N}_f, \widehat{N}_s, \widehat{\tau}_c), \quad (25)$$

where $\widehat{K}$ follows the capital-accumulation equation and $\widehat{\tau}_c$ solves the consolidated budget. Damped updates continue until capital, both labor aggregates, the consumption tax, and the budget residual satisfy tolerance.

The implementation first solves 101 types and uses that state to initialize 501 types. Validation requires positive consumption, nonnegative saving, normalized regime shares, positive masses in all three choices, two interior roots, the sequence $I \rightarrow C \rightarrow P$, the income identity, and a maximum fixed-point residual below 2×10^{-6} . The sensitivity equilibria repeat the complete fixed point rather than holding prices or taxes fixed.

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